

Supermassive Black Hole Effects on Galaxy Properties

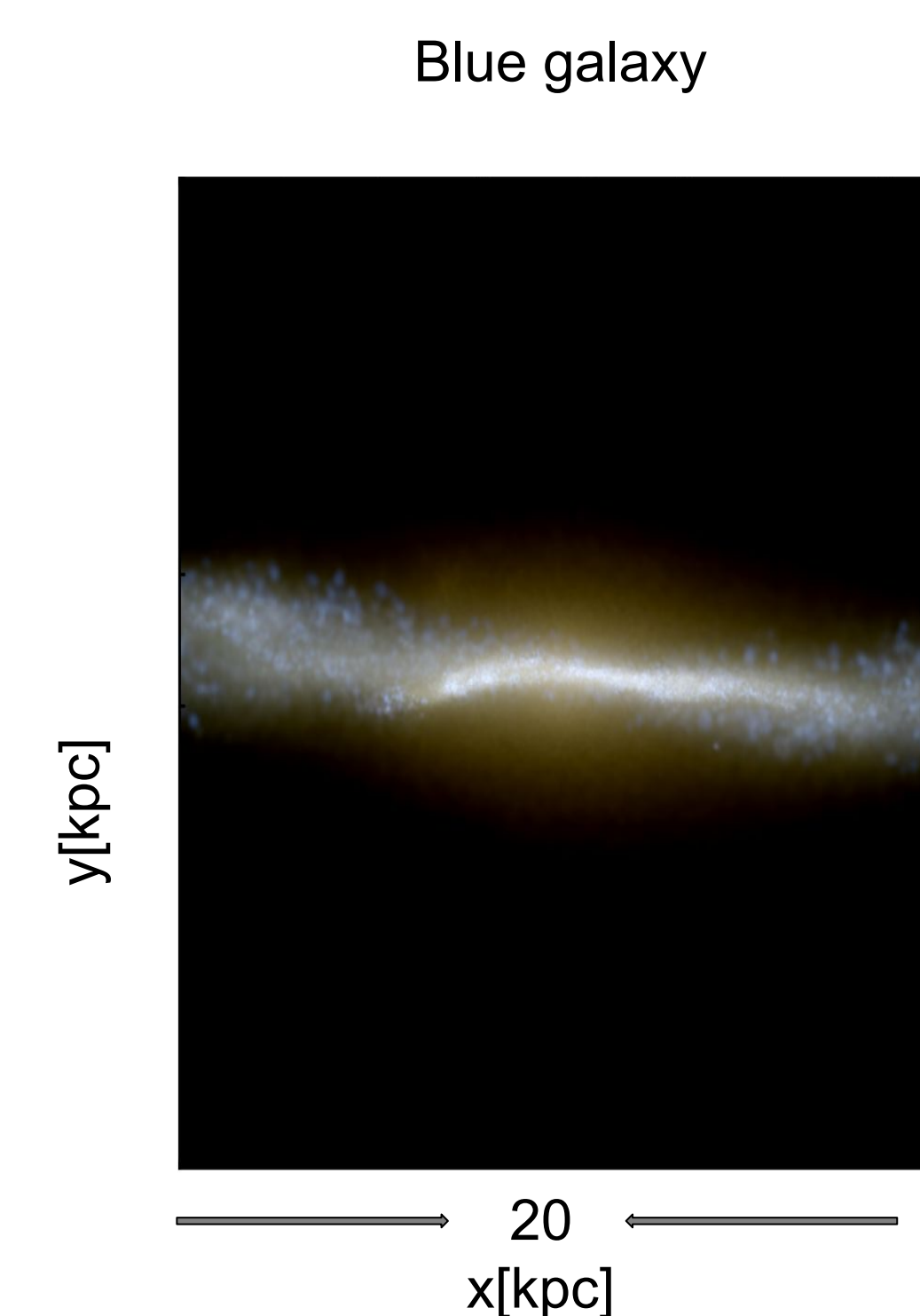
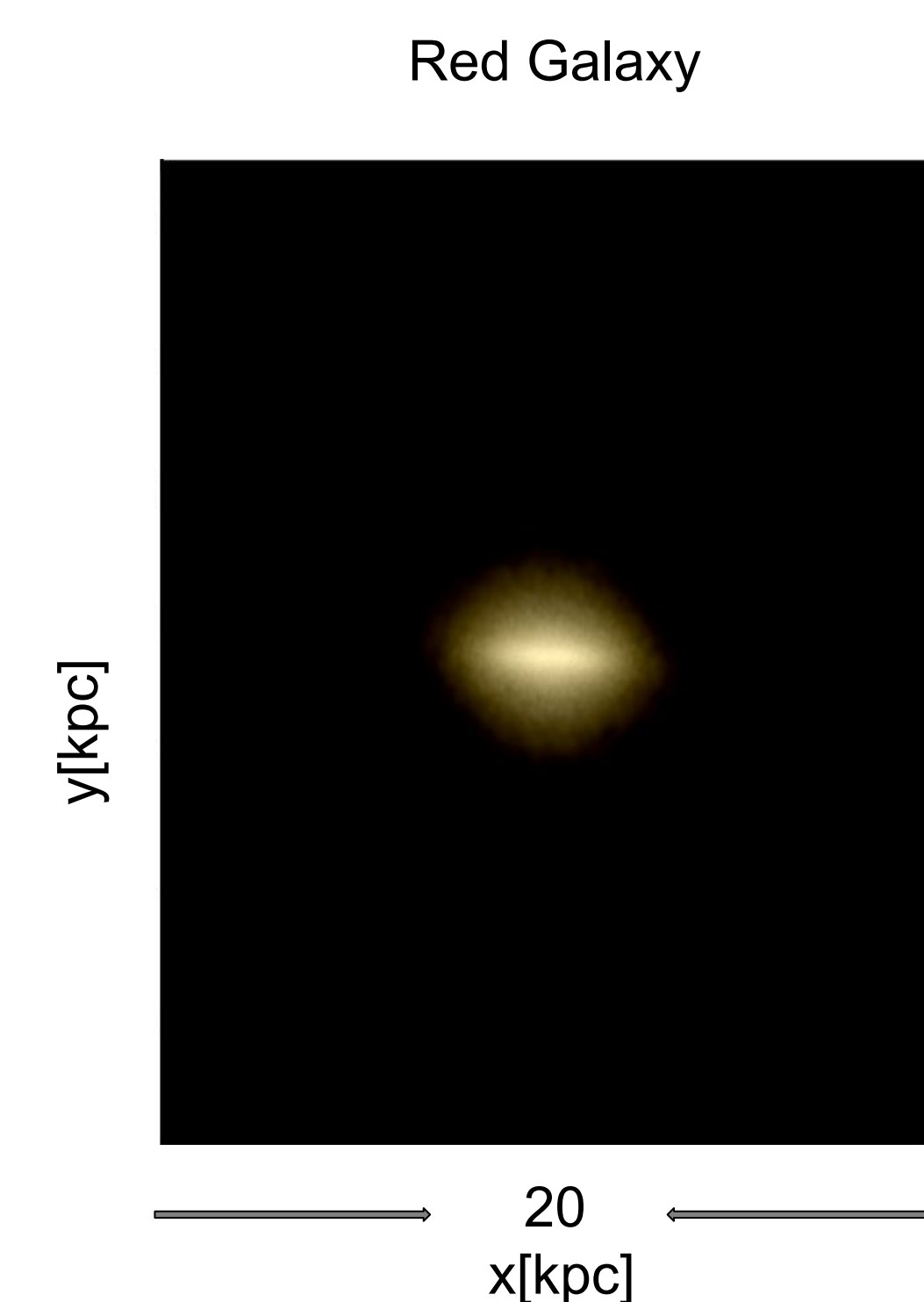
Zoe DeFord¹, Josh Smith¹

University of Washington

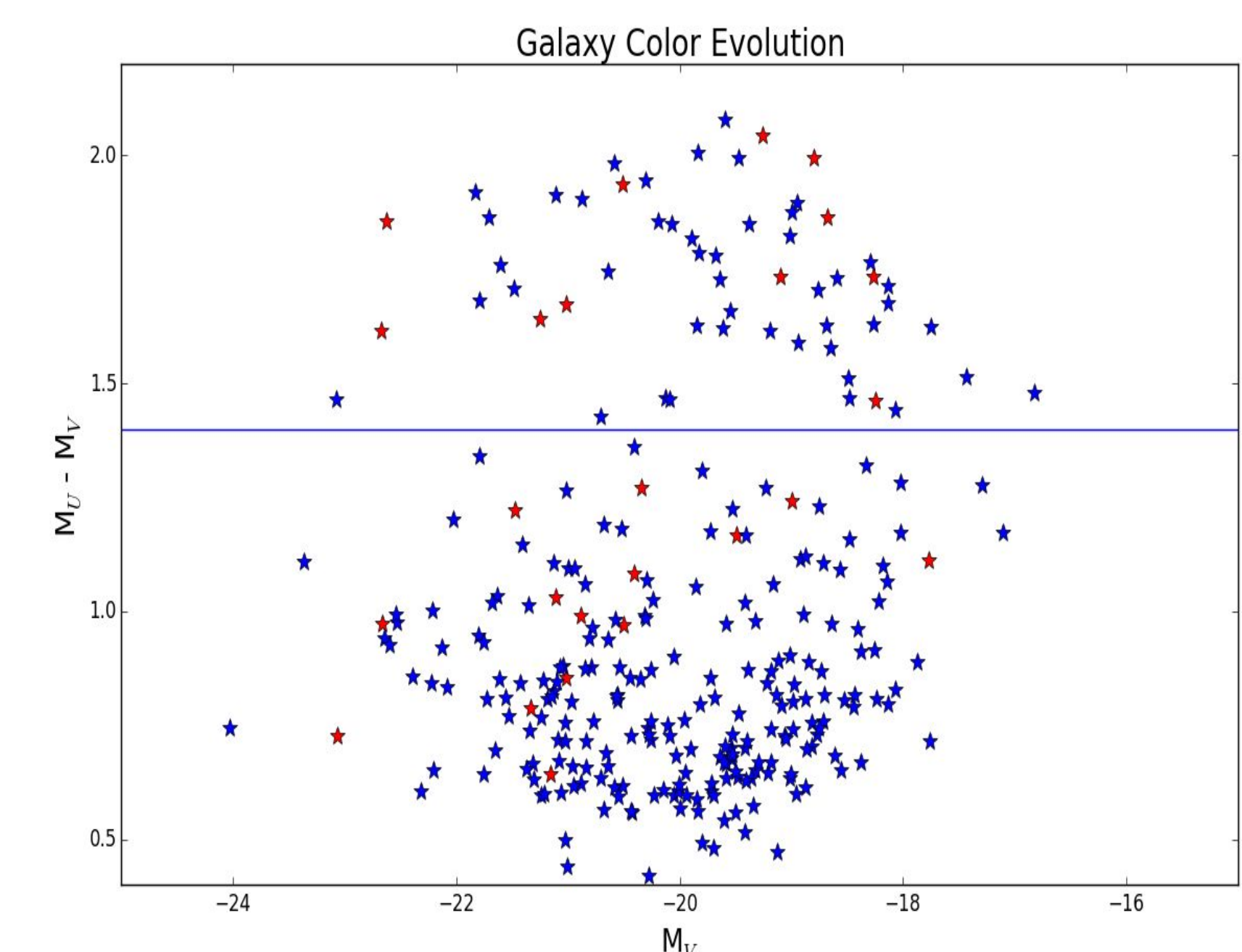
In collaboration with Professor Tom Quinn and Michael Tremmel

Abstract

Little is understood about how Supermassive black holes affect their host galaxies in terms of growth and shape. Using the Simulation Romulus25, we are able to study how galaxy properties correlate with black hole properties throughout time. The simulation is a cube with lengths of 25 Mpc on each side, which is about 75 million light years. From this simulation we drew a sample of 500 galaxies and analyzed their properties to understand how they are affected by the growth of supermassive black holes.

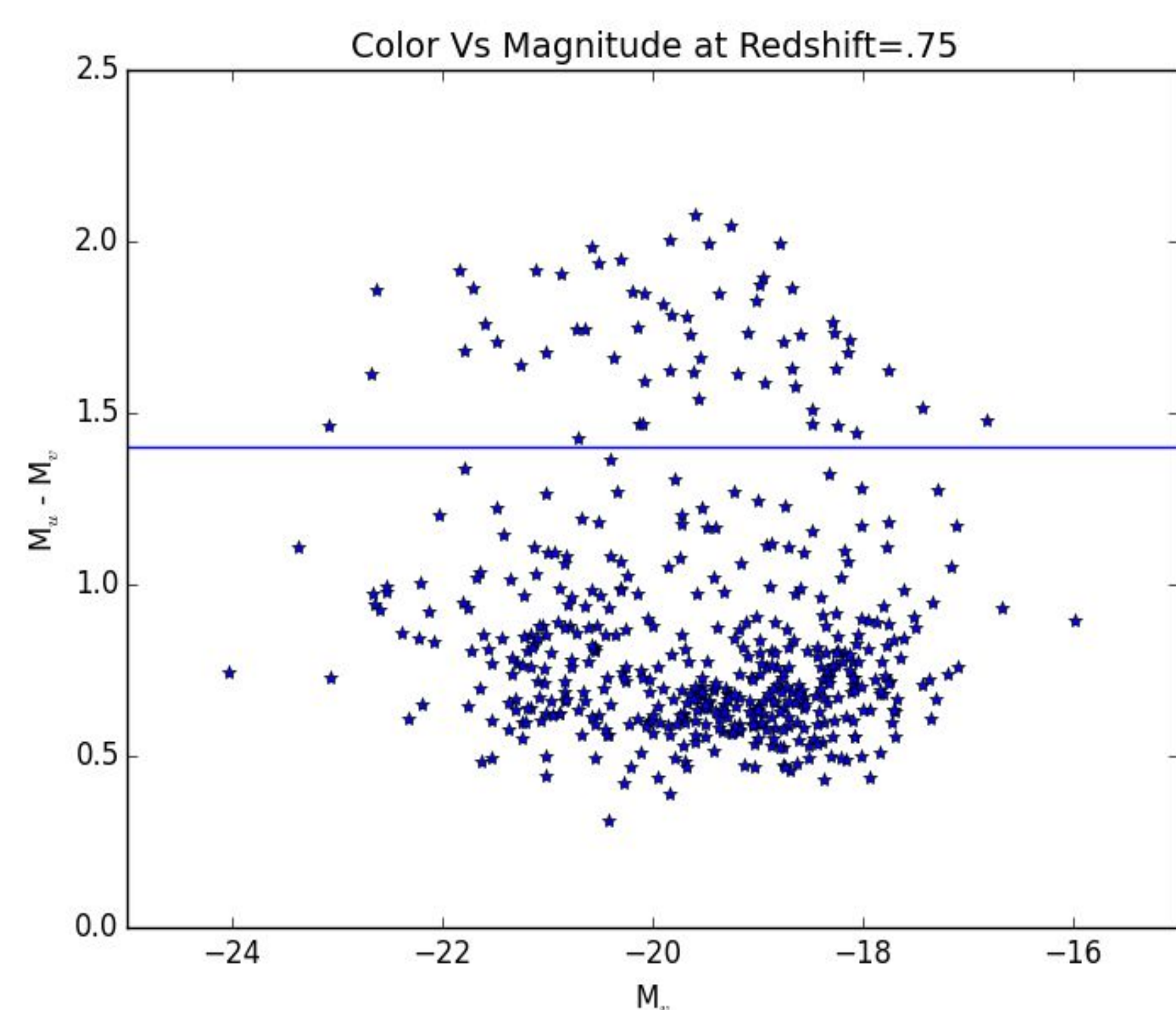
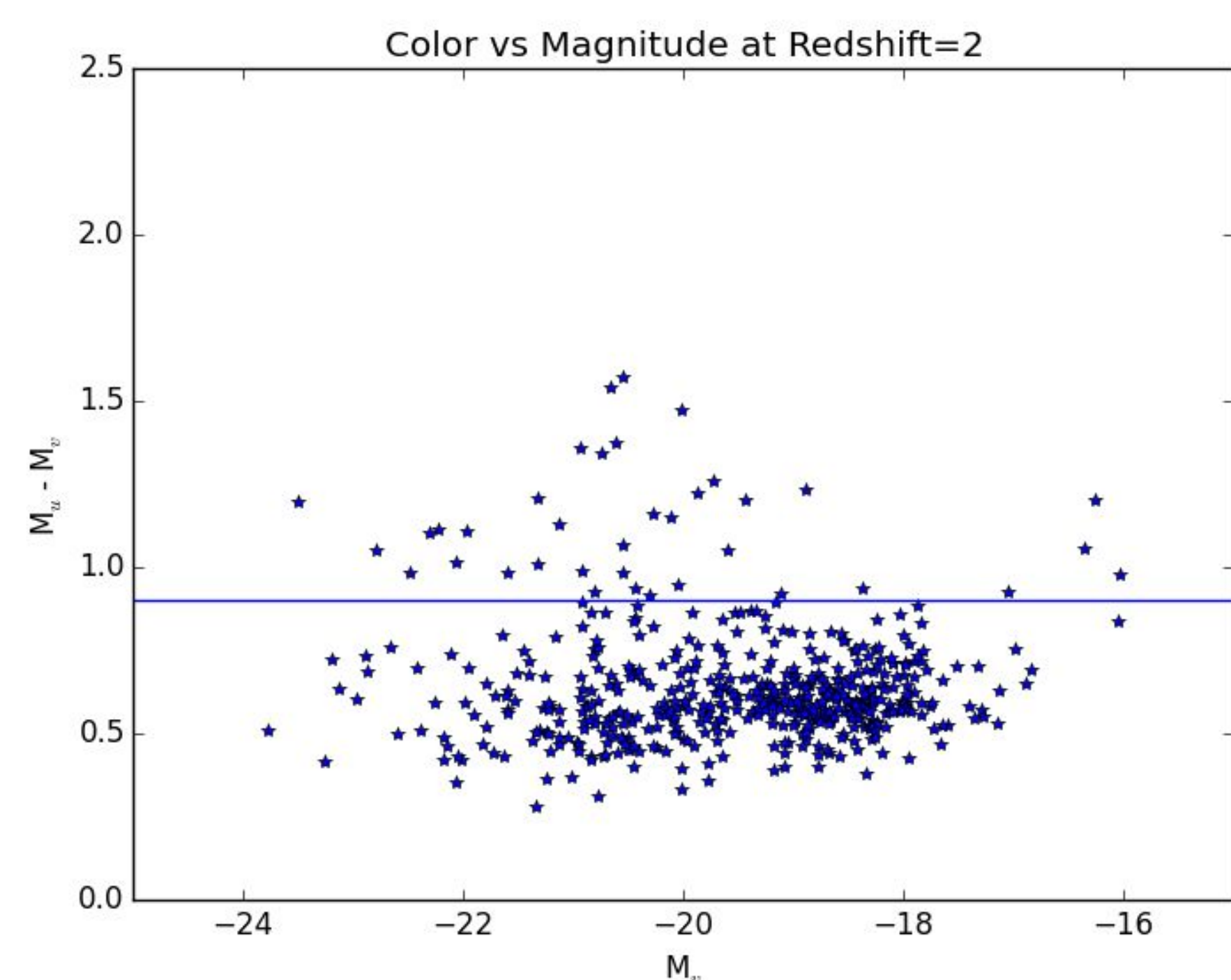


Color Evolution



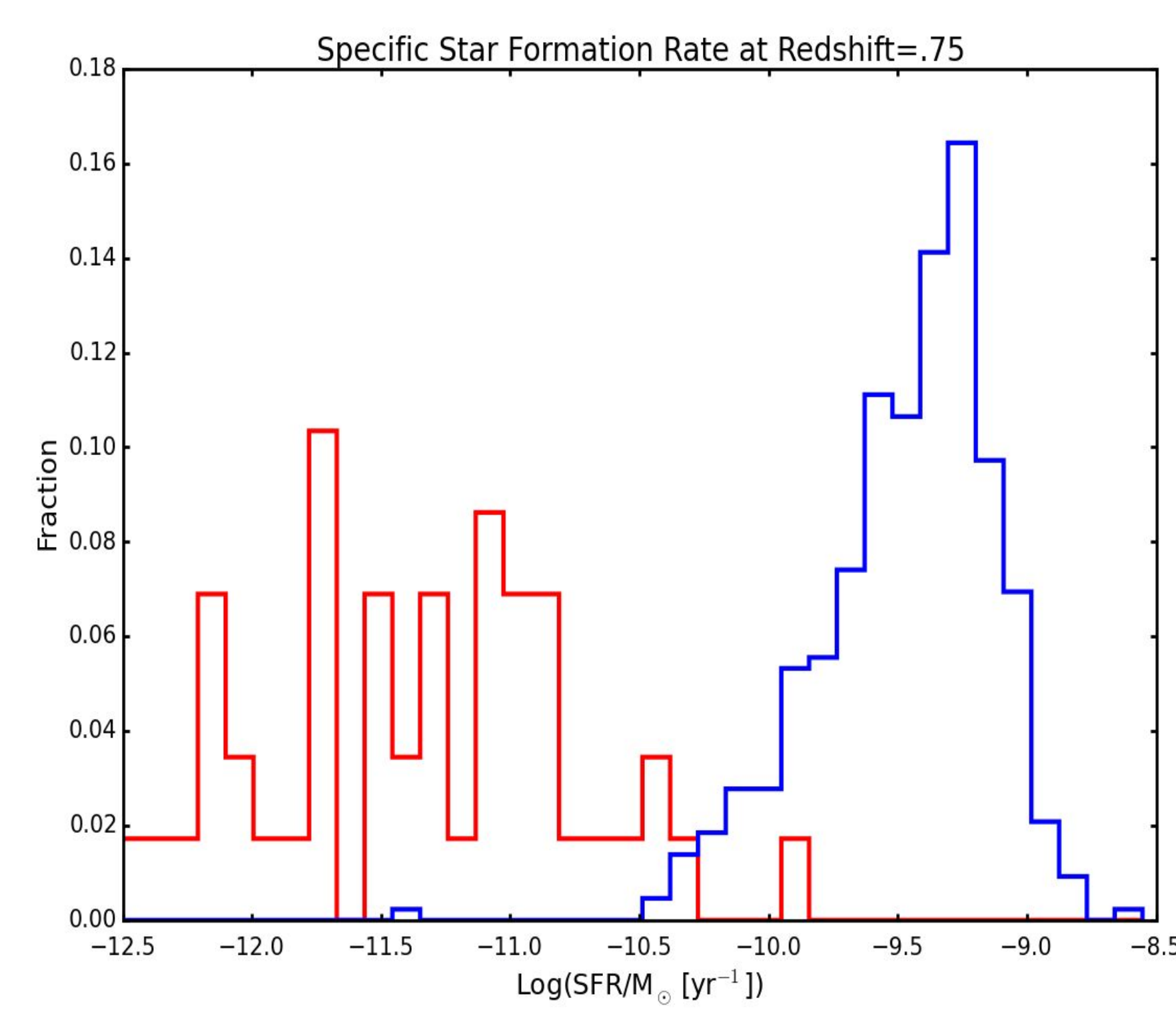
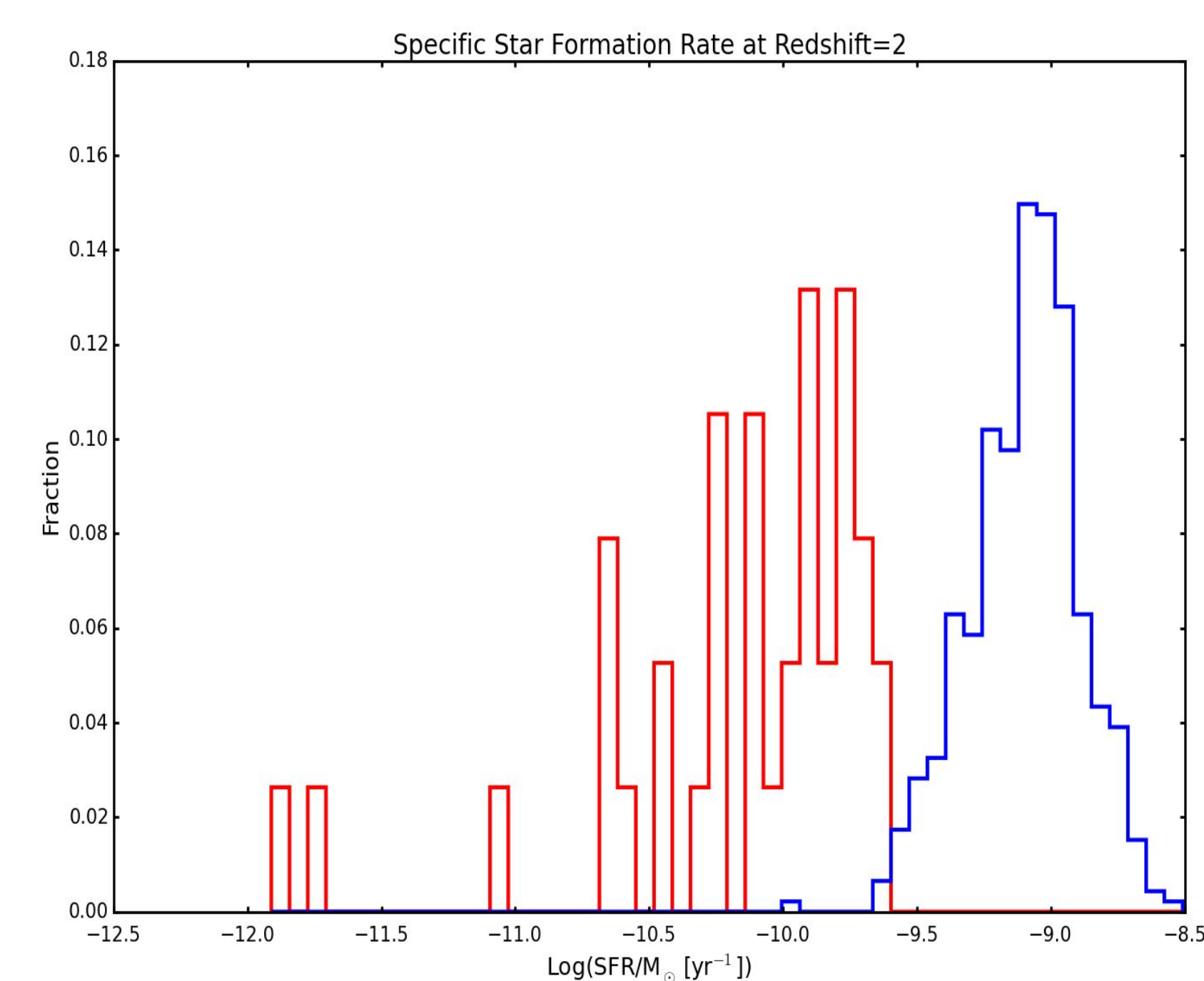
The change of galaxy colors overtime. The $z=.75$ colors and magnitudes are plotted with the $z=.75$ threshold. Galaxies that were red at $z=2$ are plotted with red stars, and galaxies that were blue are plotted with blue stars.

Color-Magnitude Bi-modality



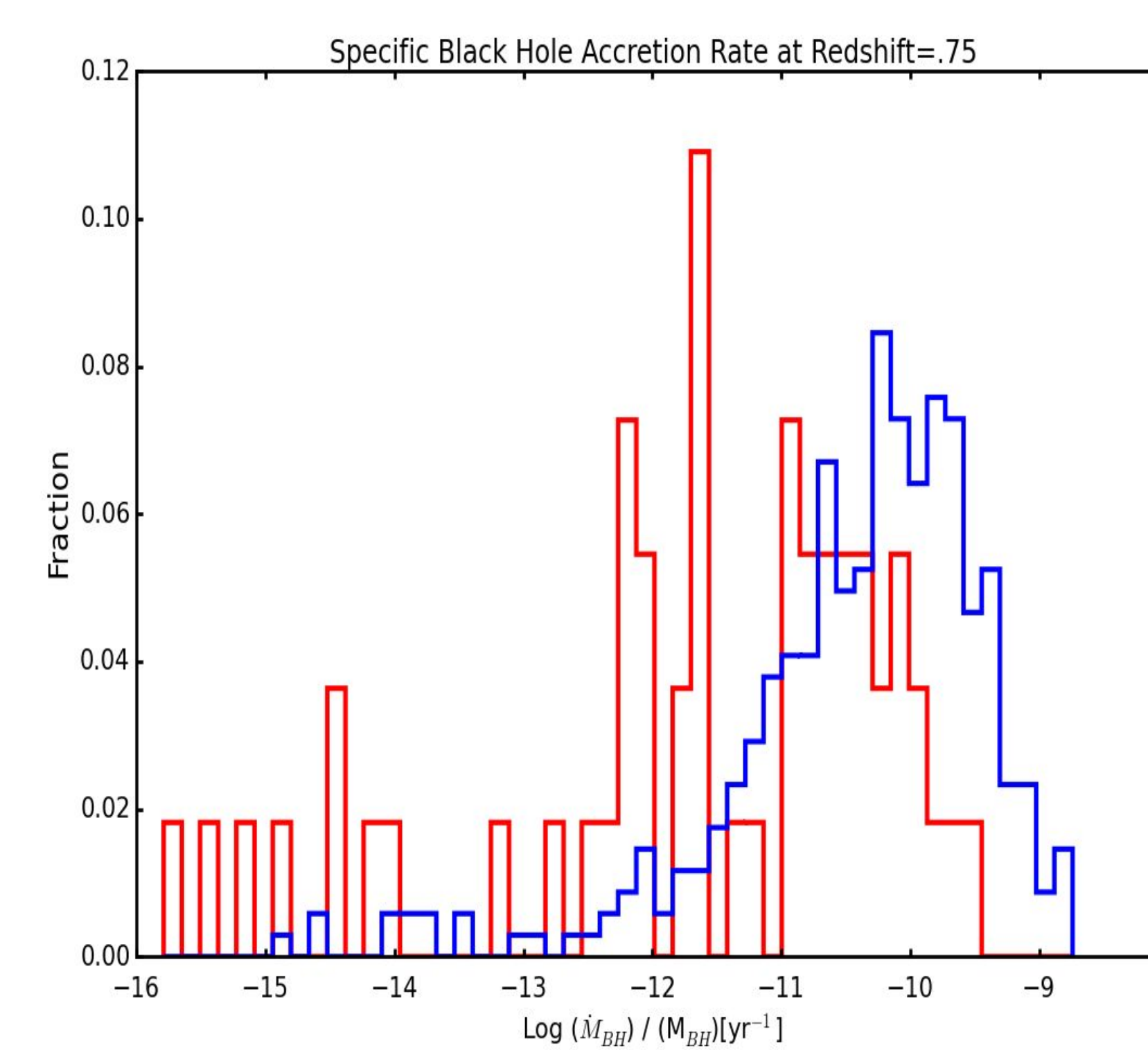
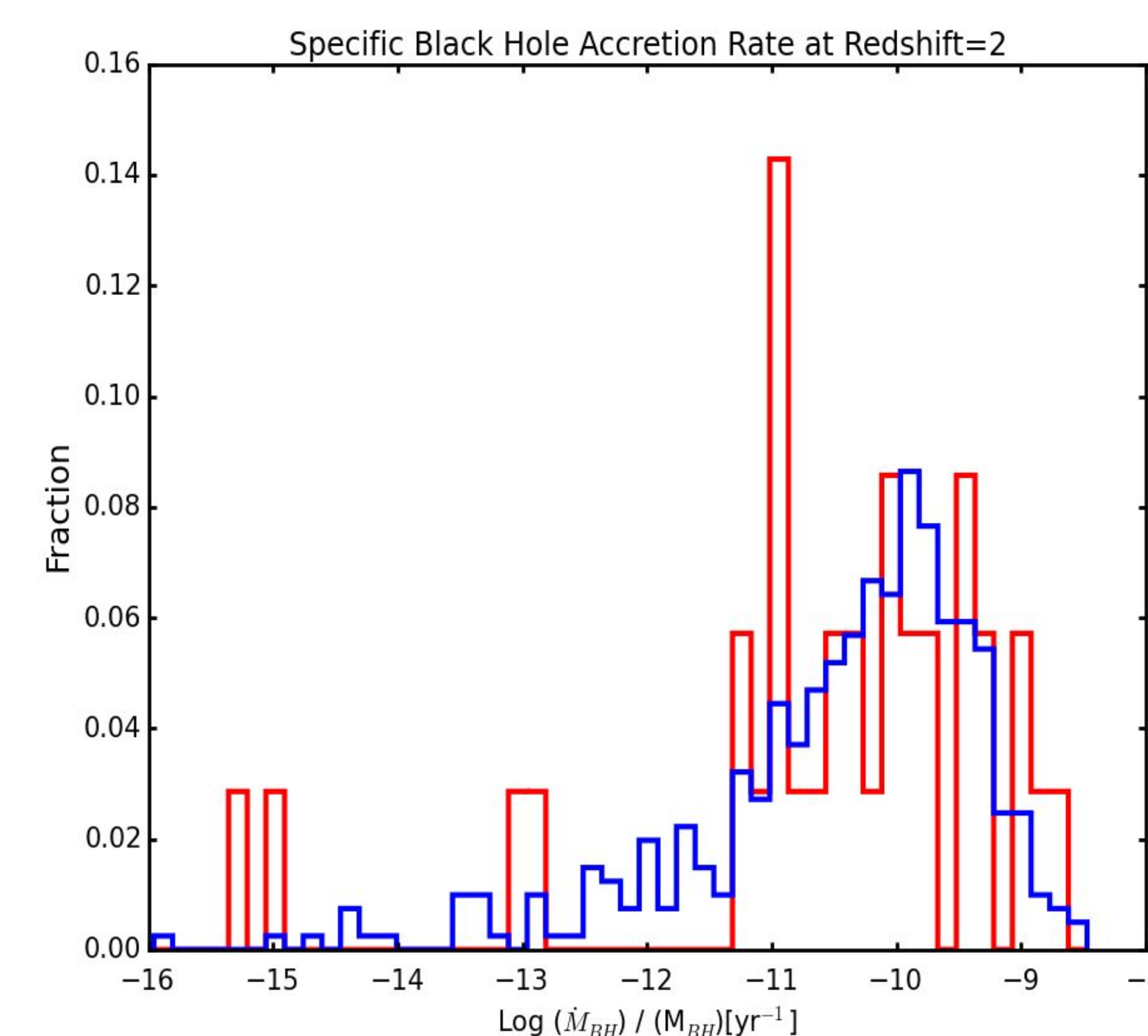
Threshold line between “red” and “blue” stars for $z=2$ (upper) and $z=.75$ (lower) The galaxies above the horizontal are classified as red, and galaxies below the line as blue.

Specific Star Formation Rate



The rate at which galaxies are forming stars divided by the stellar mass. Color corresponds to red and blue classification. Blue galaxies have higher star formation rates per unit mass.

Black Hole Growth in Red and Blue Galaxies Across Cosmic Time



There is evidence that black hole activity in red galaxies. At early times black holes are growing rapidly in both blue and red galaxies, but there is evidence that at later times black hole growth is shutting off in red galaxies while it continues in blue galaxies.

Conclusion

We are able to see that galaxies can be cleanly separated into red or blue galaxies—just like in the real universe, which leads us to observe that color distinguishes between high and low star formation rates. It is an interesting and relevant point that there is no clear correlation between color and black hole activity because contrary to expectations of previous research, black hole activity does not completely shut off star formation. In the early universe, red or blue galaxies could have high or low black hole activity, but later on ($z=.75$), red galaxies seem to have slightly less black hole activity. Galaxy color evolution shows us that most stars in galaxies follow the blue to red process of aging, but there are some intriguing red to blue galaxies that will be important in understanding the process of rejuvenation for future work.